3G TECHNOLOGY

3G refers to the third generation of developments in wireless technology. It is a collective term for the new communication procedures, standards, and devices that will improve the speed and quality of services on mobiles. 3G-compatible handsets combine the functionality of a mobile phone with that of a PC and a personal organiser/PDA.

3G divides each call or transmission into little packets of data, marking each one with an individual code to show which connection it belongs to. This is a more efficient way of transmitting data, allowing 3G networks to deliver larger files, like pictures and video, at much faster speeds.

3G devices have greater transmission abilities, both in terms of speed and capacity, than 2G or 2.5G. The International Telecommunications Union (ITU) *defines* 3G as any device that can transmit and receive data at 144Kbps or more. In practice, 3G devices can transfer data at up to 384Kbps.

Besides phone calls, 3G allows fax transmissions, e-mails, including large attachments, while on the move. High-speed internet access allows web browsing and fast downloading of data files, software, image or music files. 3G can be used for video conferencing and some 3G handsets can also function as personal organisers, with electronic diaries, contact lists, and automatic reminders. Most 3G networks offer global roaming.

Japan is the *first* country which introduced 3G on a large commercial scale in 2005. 3G is now also in use in France, Germany, and Austria besides some other countries.

There is no single global 3G standard, but the principal technologies of 3G include:

(i) WCDMA, which has been chosen for 3G mobile phone systems in Europe, Asia, and the US. It first converts raw data into a narrow band digital radio signal and then attaches a marker to each data packet to identify it as belonging to a particular communication.

(ii) Customers who already use CDMA can upgrade to newer models. CDMA2000 IxEV-DO provides always–on packet data connection like landline-based broadband, for mobile internet use.